



FEB 9 – MODEL COMPARISON + SIMPLE TASKS

ACTIVITY I

- Dataset: Weather AUS
- <https://www.kaggle.com/code/drisharahmad/australia-weather-forecast>
- Let's read the description, and pretend we don't have LLMs. What kind of things would we like to do or know?
- Are there missing data?
- Look through the columns and figure out if some should be discarded as obviously not useful
- What are the data sources/How recent is the data set?
- Ranges, outliers and anomalies, balance of target variable
- Basic visualization: graphing of the variables
- Correlations
- Prompt: Analyze the data file "weatherAUS.csv" from the "Rain in Australia" dataset on Kaggle. What are some preliminary considerations about these data?
- Pick two models. As a class, we can actually run a few models!

ACTIVITY I (CONT'D)

- Look at the output from two models. We can spend some time comparing results and ranking model choices.
- Is any of this helpful? How can we modify the prompt according to our priorities?
- Let's run the updated prompt and discuss the result.
- Anybody wants to propose a new task?

ACTIVITY 2 (WE DIDN'T GET TO THIS, BUT I LEFT IT HERE)

- <https://www.kaggle.com/datasets/berkeleyearth/climate-change-earth-surface-temperature-data>
- Let's read the description together.
- Suggested prompt: Using the Berkeley Earth Surface Temperature dataset, write code using Plotly to create an interactive line chart. I want to visualize the temperature anomalies for 'London', 'New York', and 'Tokyo' from 1900 to 2013, with a rolling 12-month average to smooth out the seasonal noise.
- On your own:
- Develop a new plotting request that is more closely related to climate change discovery.
- Ask to make a table of some relevant results and format it in LaTeX.
- What other tasks would you like to perform?

HOMEWORK

Think about a task (data exploration, visualization, data imputation, formatting...) that could be useful for your research. Keep it simple – we'll have specialized sessions for coding, literature search, improving research ideas, etc.

You can pick one or two models (consider using arena.ai or something else that offers more privacy).

We'll have a co-working session on Wednesday and get it started, or even better, done!